# Refinement action: cut energy, quantum phase and Gaussian information

Maintained result. For a constant force $F \neq 0$, mass $m > 0$ and duration $\tau > 0$, the chord/parabola energy is $K_\tau = F^2\tau^3/(24m)$. An inserted cut at fraction $s$ spends $3s(1-s)K_\tau$; orthogonal hat energies add and exhaust $K_\tau$ exactly when the mesh vanishes. Three uses of that same functional have different assumptions:

- Cut geometry is a classical energy identity and selects no positive action.
- Supplied Weyl relations convert remaining polygon energy into a state-independent relative phase $\theta_N = F^2\sum_j \tau_j^3/(24m\hbar)$. It is observable through a specified coherent control; repeated passes change its resolution threshold.
- A prescribed Euclidean Gaussian bridge gives KL information $K_\tau/\hbar$. Independent uncorrelated Gaussian position marks with a supplied product floor $\kappa$ have sharp finite-grid information $(K_\tau - \sum_j K_{\tau_j})/\kappa$, under the explicit availability assumptions below. These are not arbitrary quantum-instrument bounds.

The zero-force case has zero energy at every mesh; the mesh/exhaustion equivalence then has no content. The general-force extension below uses the Dirichlet Green functional; its mark bound is attainable in the specified dense construction for forces of one sign, not automatically for sign-changing forces.

The calculations use standard Schauder/Cameron–Martin, Weyl and linear Gaussian methods. Source passages and historical interpretations are identified where used. Internal Astra/Fable reviews checked written formulas and prompted scope corrections; they are not independent peer review or literature novelty evidence. The insertion treatment supplies the three-area source geometry and a distinct controlled-loop experiment.

## 1. One cut

Put $C = F^2/(24m)$, so $K_\tau = C\tau^3$. On $[0,\tau]$ the parabola with acceleration $F/m$ and the chord through its endpoints differ by

$$\delta x(t) = \frac{F}{2m}\,t(\tau - t), \qquad \dot{\delta x}(t) = \frac{F}{m}\Big(\frac{\tau}{2} - t\Big),$$

so $\frac{m}{2}\int_0^\tau \dot{\delta x}^2 dt = \frac{m}{2}\cdot\frac{F^2}{m^2}\cdot\frac{\tau^3}{12} = C\tau^3$.

Cut at $t_c = s\tau$. The polygon through the three points $0, s\tau, \tau$ differs from the chord by the hat $\phi$ of height $h = \delta x(s\tau) = Fs(1-s)\tau^2/(2m)$, linear on $[0, s\tau]$ and on $[s\tau, \tau]$. The polygon differs from the parabola, on each piece, by the chord-minus- parabola of that piece: $\delta x = \phi + \delta x_1 + \delta x_2$ with $\delta x_j$ supported on piece $j$ and vanishing at its ends.

Orthogonality. $\dot\phi$ is constant on each piece and $\int_{\text{piece}} \dot{\delta x}_j = 0$, so $\int \dot\phi\,\dot{\delta x}_j = 0$; the $\delta x_j$ have disjoint supports. Hence

$$K_\tau = \frac{m}{2}\int \dot\phi^2 + K_{s\tau} + K_{(1-s)\tau}.$$

The hat' s energy. $\frac{m}{2}\int\dot\phi^2 = \frac{m}{2}h^2\big(\frac{1}{s\tau} + \frac{1}{(1-s)\tau}\big) = \frac{mh^2}{2s(1-s)\tau} = \frac{F^2 s(1-s)\tau^3}{8m} = 3s(1-s)C\tau^3$. The same number follows from $\tau^3 = (s\tau)^3 + ((1-s)\tau)^3 + 3s(1-s)\tau^3$, which is the algebraic form of Theorem 1. The share is largest for halving, $\frac{3}{4}K_\tau$. □

## 2. Any cut sequence

Each later cut acts inside one current piece, and Theorem 1 applies to that piece with its own $s$. The new hat is supported in the piece and vanishes at its ends; every earlier hat is linear there. The one-cut argument gives orthogonality to all earlier hats and to the other pieces' remainders. After any finite set of cuts, with pieces $\tau_i$ ($\sum \tau_i = \tau$),

$$K_\tau = \sum_{\text{cuts}} \frac{m}{2} \int \dot{\phi}^2_{\text{cut}} + \sum_i K_{\tau_i}, \qquad \sum_i K_{\tau_i} = C \sum_i \tau_i^3 \le C\tau \max_i \tau_i^2.$$

For $F \neq 0$, the remainder vanishes iff the mesh does ($\sum_i \tau_i^3 \ge (\max_i \tau_i)^3$). The positions and the order enter only through which pieces are cut, never through the total. □

This is the Schauder (Lévy–Ciesielski) expansion of $\delta x$ in the Cameron–Martin space $H^1_0[0,\tau]$, adapted to an arbitrary nested cut sequence; for dyadic cuts it is the textbook expansion (Ciesielski 1961; metadata). The reachability note has the same cubic law for the area $2F^2t^3/(3m)$ of the reachable lens of a cut. The bridge's own conditional variance shows the same nested additivity with a quadratic law: for diffusion constant $D$, $\int_0^L \mathrm{Var}(X_u \mid \text{ends})\, du = DL^2/6$, and a cut at fraction $s$ removes $Ds(1-s)L^2/3$ of it.

## Historical cut schedules

Zhuangzi 33 (dialecticians' list, text): 一尺之捶，日取其半，萬世不竭，"a one-foot stick, each day take half, in ten thousand generations it is not exhausted." Mohist Canon B (B60 in Graham's numbering; canon): 非半弗斮，則不動，說在端，"without halving there is no cutting and no moving; the explanation lies in the point (端)", with the explanation (text) 前則中無爲半，猶端也。前後取，則端中也 (working gloss: "going forward, at the middle nothing serves as a half: it is like a point; taking from front and back, the point is in the middle"), which continues 斮必半，"cutting must be by halves".

One-ended schedule. Day $k$ halves the remaining piece, of length $\tau 2^{-(k-1)}$, and spends $\frac{3}{4}C\tau^3 8^{-(k-1)}$. The total is $\frac{3}{4}\cdot\frac{8}{7}K_\tau = \frac{6}{7}K_\tau$; the pieces taken away keep $\sum_{k\ge1} 8^{-k}K_\tau = \frac{1}{7}K_\tau$.

Alternating schedule (the reading favoured by 斮必半). One halving a day, taking the half alternately from the front and from the back. Each day is one halving of the remaining piece, so the spending is that of the one-ended schedule, $\frac{6}{7}K_\tau$; the remainder converges to the point $\frac{\tau}{2} + \frac{\tau}{8} + \cdots = \frac{2\tau}{3}$.

Two quarter-cuts a day (a construction motivated by 前後取; the text does not fix the fractions). Day $k$ cuts the remaining piece $r = \tau 2^{-(k-1)}$ at $s = \frac{1}{4}$ and $s = \frac{3}{4}$, keeping the middle half. The two cuts spend $Cr^3\big(1 - 2\cdot\frac{1}{64} - \frac{1}{8}\big) = \frac{27}{32}Cr^3$ (equivalently $\frac{9}{16} + \frac{9}{32}$ by two applications of Theorem 1), and the total is $\frac{27}{32}\cdot\frac{8}{7}K_\tau = \frac{27}{28}K_\tau$, with $\frac{1}{28}K_\tau$ kept in the pieces taken away. The contrast $\frac{6}{7}$ against $\frac{27}{28}$ is between one half-cut and two quarter-cuts a day, whichever ends are cut.

All three schedules converge to a point, the Mohist 端: the end, the point $2\tau/3$, and the midpoint. If 中 in 端中 means the midpoint, the explanation fits the symmetric schedule, while 斮必半 fits the alternating one. What each author is committed to, and what the theorem does with it:

- The dialecticians hold that the halving never ends. Theorem 2 agrees for the figure and adds a price: their schedule spends only $\frac{6}{7}$ of the action, since it never cuts again what it took away.

- The Mohists hold that the halving ends at a point. Theorem 2 agrees that each schedule converges to a point, and the point carries no action. The unhalvable 端 as a least magnitude is a reading the theorem leaves aside, as the Planck paper already records.
- Newton's scholium closing Book I, Section I, takes the geometers' side (divisibility without end). Theorem 2 is that side made quantitative: refinement exhausts the action only when every piece is cut again.

The parallel with Zeno's dichotomy (Aristotle, Physics VI.9) is recorded as convergence; transmission would need evidence this note does not have.

## Gaussian bridge information

Take the Euclidean path measure of a free particle with variance rate $\hbar/m$: unconditioned centred Brownian motion $X$ with $E[X_t^2] = \hbar t/m$, then pin it at the cell's endpoints. Given the endpoints of a piece of length $L$, the value at fraction $s$ is Gaussian with variance $\hbar s(1-s)L/m$, independent of the finer hats and of the other pieces, by the Markov property of the bridge (the Lévy construction, for any nested sequence). A force $F$ shifts the mean of that value by the sagitta $h = Fs(1-s)L^2/(2m)$. The Kullback–Leibler divergence of the shifted from the free law (the expected log-likelihood ratio, which here also equals the ratio evaluated at the shift) is

$$\frac{h^2}{2\hbar s(1-s)L/m} = \frac{mh^2}{2s(1-s)L\hbar} = \frac{3s(1-s)CL^3}{\hbar},$$

the spent share of Theorem 1 divided by $\hbar$. By the chain rule the divergences add over cuts, and along a sequence with mesh tending to zero they sum to $K_\tau/\hbar$, the Cameron–Martin exponent for the shift $\delta x$. The Cameron–Martin density factorizes over cuts, each contributing the normalizing factor $e^{-3s(1-s)CL^3/\hbar}$. This is the Euclidean counterpart of the real-time two-path phase $K_\tau/\hbar$ of the two-path note (with the physical phase unit $\hbar$ supplied), and of the product-over-cuts structure that § 7 of the fifth-postulate note reads in the time-step partition of Rivero 1998. For $\hbar \to 0$ every cut, however small, is decisive; for $\hbar > 0$ the cuts whose share falls below $\hbar$ carry little evidence (under one nat). □

## Conditional cut counts

If every cut that a record exhibits must spend at least $\kappa > 0$, Theorem 2 gives at most $K_\tau/\kappa$ such cuts, for every schedule. For halving refinement the $2^n$ cuts of level $n$ each spend $\frac{3}{4}K_\tau 8^{-n}$, so the exhibited levels are those with $8^n \le 3K_\tau/(4\kappa)$.

Two floors must be kept apart: $\kappa$ here floors the share a cut spends, $3s(1-s)K_L$, while the thresholds in the other notes floor the action of a cell, $K_\tau$; for halving the two differ by the factor $\frac{3}{4}$. The Planck paper's eq. (2) is $K_\tau \ge 2z_{1-\epsilon}^2\hbar$: $\tau_*$ is the cell with $K_{\tau_*} = 2z_{1-\epsilon}^2\hbar$, whose halving spends $\frac{3}{2}z_{1-\epsilon}^2\hbar$. The round-3 resolution threshold $\tau \ge (24m\hbar d_{n,\eta}/F^2)^{1/3}$ is $K_\tau \ge \hbar d_{n,\eta}$, that is $\kappa = \frac{3}{4}\hbar d_{n,\eta}$ in cut-share form. Corollary 5 thus places a floor, when one is given, at a definite depth of the cut process; it supplies no floor.

## Full-path Gaussian testing

Keep the measure of § 4: unconditioned centred Brownian motion with $E[X_t^2] = \hbar t/m$, the Euclidean free-particle measure, whose diffusion coefficient $\hbar/2m$ is the one of Nelson (1966) (metadata). Assume $m, \tau, \hbar > 0$, $F \in \mathbb{R}$, equal priors and $0 < \epsilon < 1/2$. Let $\ell$

be a fixed endpoint chord and let $B$ be the centred bridge with covariance $E[B_u B_v] = (\hbar/m)(\min(u,v) - uv/\tau)$. Compare $P_0 = \mathcal{L}(\ell + B)$ and $P_1 = \mathcal{L}(\ell - \delta x + B)$, where $\delta x(u) = Fu(\tau - u)/(2m)$. This specifies additive position noise around the two Newtonian comparison paths. Their Cameron–Martin norm is $d^2 = \|\delta x\|^2_{\mathrm{CM}} = (m/\hbar)\int \dot{\delta x}^2 = 2K_\tau/\hbar$. Writing $h = -\delta x$, the log-likelihood ratio is $Y - d^2/2$, where $Y = \langle h, X - \ell\rangle_{\mathrm{CM}}$ denotes the Gaussian stochastic pairing, defined even though typical paths lie outside the Cameron–Martin space. Here integration by parts makes it the ordinary area statistic $Y = -(F/\hbar)\int_0^\tau (X_u - \ell_u)du$. Under $P_0$, $Y \sim N(0, d^2)$; under $P_1$, $Y \sim N(d^2, d^2)$. Thus the Neyman–Pearson test with equal priors thresholds that statistic halfway and errs with probability

$$p_{\mathrm{err}} = \Phi\Big(-\tfrac{1}{2}\|\delta x\|\Big) = \Phi\Big(-\sqrt{K_\tau/2\hbar}\Big).$$

Hence $p_{\mathrm{err}} \le \epsilon$ iff $K_\tau \ge 2z^2_{1-\epsilon}\hbar$, that is $\tau\Delta E = 12K_\tau \ge 24 z^2_{1-\epsilon}\hbar$ and, for $F \neq 0$, $\tau \ge \tau_* = (48 z^2_{1-\epsilon} m\hbar/F^2)^{1/3}$: eq. (2) of the Planck paper, including its "about $65\hbar$ at $\epsilon = 0.05$" ($2z^2_{0.95} = 5.41$). A single cut observed alone gives $\Phi(-\sqrt{\mathrm{share}/2\hbar})$ in the same way. □

The Planck paper reaches (2) from real-time Gaussian marks whose record and recoil obey Robertson's bound $\hbar/2$; here the same constant comes from the Euclidean path measure $e^{-S/\hbar}$ through the Cameron–Martin theorem alone. The agreement is exact and structural. In Theorem 2 of the Planck paper, the mark trade-off $\kappa_{\mathrm{mark}} = \hbar/2$ gives $\sup d^2_{\mathrm{mark}} = F^2\tau^3/(24m\kappa_{\mathrm{mark}}) = 2K_\tau/\hbar$. Its optimization uses the same Dirichlet inverse of $-d^2/du^2$ with zero boundary values, hence the same parabolic extremizer and $\tau^3/12$. The full bridge attains this Gaussian information; the independent, uncorrelated, non-adaptive mark protocols approach it in the dense limit for tests invariant under initial position and velocity. This equality of sharp information retains the different experimental premises; it supplies no equivalence of all records or of Nelson's state-dependent dynamics. Both routes supply $\hbar > 0$, giving a second conditional derivation of the mesh, with the necessity question of the fifth-postulate note unchanged. What it adds is the reading of the floor on recording the inertial–parabola difference as the distinguishability threshold of the force in the free-particle path measure. At small $\epsilon$, $z^2_{1-\epsilon} \simeq 2\ln(1/\epsilon)$, so this floor grows like $4\hbar\ln(1/\epsilon)$ in cell action, the same logarithmic law as Theorem E of the fifth-postulate note, $ab \ge \hbar c_*(\epsilon)$ with $c_*(\epsilon) \simeq \frac{1}{2}\ln(1/\epsilon)$ in window half-widths; the constants refer to different quantities (a cell's Galileo action and a phase-space window).

## Finite-grid Gaussian mark information

Proposition 7. Take the mark model of the Planck paper, § 3: mark $j$ returns the position with Gaussian error $\delta_j$ and delivers a Gaussian impulse $\Delta_j$ with $\delta_j\Delta_j \ge \kappa$; protocols are non-adaptive and tests invariant under the unknown initial position and velocity, and marks with $\delta\Delta = \kappa$ are available at every $\delta > 0$ (as in the sharpness of Theorem 2; the upper bound needs only $\delta\Delta \ge \kappa$). Let the marks sit at $t_0 < t_1 < \cdots < t_N$, with $\tau = t_N - t_0$ and pieces $\tau_i = t_i - t_{i-1}$. Then the supremum over the marks' $\delta_j, \Delta_j$ of the discernibility is

$$\sup d^2 = \frac{K_\tau - \sum_i K_{\tau_i}}{\kappa} = \frac{1}{\kappa}\sum_{\mathrm{cuts}} \frac{m}{2}\int \dot{\phi}^2_{\mathrm{cut}}, \qquad K_\tau = \frac{F^2\tau^3}{24\,m},$$

the action that the interior cuts spend by Theorems 1–2. It is approached, never attained, with sharp end marks and balanced interior marks. Special cases: a single cut at fraction $s$ gives $3s(1-s)K_\tau/\kappa$; $N$ equal pieces give $(1-N^{-2})K_\tau/\kappa$, so $n$ halvings give $(1-4^{-n})K_\tau/\kappa$; the dense limit gives $K_\tau/\kappa$, the paper's Theorem 2. Adding a mark inside

a piece adds exactly that cut's share, $3s(1-s)K_{\tau_i}/\kappa$: the squared statistical distance is additive under refinement at every finite stage.

Proof. Write $T(w) = \sum_{t_i>w} u_i(t_i - w)$ for the test weights $u$. The invariance constraints $\sum u_i = \sum u_i t_i = 0$ say exactly that $T$ is piecewise linear with kinks at the $t_j$ and vanishes outside $(t_0, t_N)$, and every such $T$ arises from one $u$, with $u_j$ the jump of $T'$ at $t_j$. The paper's §3 gives the signal $\frac{F}{m}\int T$ and the noise $\sum_j \delta_j^2 u_j^2 + \sum_j \Delta_j^2 T(t_j)^2/m^2 \geq \frac{2\kappa}{m}\sum_j |u_j|\,|T(t_j)|$, by the arithmetic–geometric mean. Summation by parts gives $\sum_j u_j T(t_j) = -\int T'^2$, hence

$$d^2 \leq \frac{F^2}{2m\kappa}\,\frac{(\int T)^2}{\int T'^2}.$$

Let $V_h$ be the piecewise-linear functions on the grid vanishing at $t_0, t_N$, and $T_h \in V_h$ the Galerkin solution of $-T'' = 1$: $\int T_h'\varphi' = \int \varphi$ for all $\varphi \in V_h$. Cauchy–Schwarz in the energy product gives $(\int T)^2 \leq \int T_h'^2 \int T'^2$ with $\int T_h'^2 = \int T_h$, so the ratio is at most $\int T_h$. In one dimension $T_h$ interpolates $p(w) = \frac{1}{2}(w-t_0)(t_N-w)$ at the nodes, since the Dirichlet Green's function with its source at a node lies in $V_h$ and the error $p - T_h$ is Galerkin-orthogonal to $V_h$. So $\int T_h$ is the trapezoid rule for $p$ (nodal exactness of one-dimensional linear elements: Strang and Fix, An Analysis of the Finite Element Method, 1973; recalled), whose error on a piece is $-\tau_i^3 p''/12 = \tau_i^3/12$, and $\int T_h = (\tau^3 - \sum_i \tau_i^3)/12$. This gives the upper bound $F^2(\tau^3 - \sum \tau_i^3)/(24m\kappa)$. For the supremum take $T = T_h$, which is nonnegative and concave, so all $u_j T(t_j)$ have one sign; choose $\delta_j = \Delta_j T_h(t_j)/(m|u_j|)$ with $\delta_j\Delta_j = \kappa$ at interior marks, and let the end marks be sharp, $\delta_0, \delta_N \to 0$ with $\Delta_0, \Delta_N \to \infty$. Their impulses drop out of every invariant statistic, since $S_0 = S_N = 0$: the one at $t_N$ reaches no later reading, and the one at $t_0$ acts as a shift of the unknown $v_0$. The value is never attained, because $u_0, u_N \neq 0$ leave $\delta_0^2 u_0^2 + \delta_N^2 u_N^2 > 0$ at any finite end mark. Equivalently, $(F/m)T_h$ is the energy projection of the Galilean displacement onto the polygons on the grid, so $\sup d^2 = \frac{1}{\kappa}\frac{m}{2}\int(\Pi_h \delta x)'^2$, which is Theorems 1–2 by construction. □

The two constructions of this note and the paper's are therefore one: the Newton cell action spent by any finite set of cuts is, over $\kappa$, the best discernibility of a mark protocol placed at those cuts. With $\kappa = \hbar/2$ the dense value reproduces Proposition 6's exponent, $\frac{1}{2}\sqrt{K_\tau/\kappa} = \sqrt{K_\tau/(2\hbar)}$, from the mark model instead of the path measure. The continuity note reads the additivity as Leibniz's petites perceptions.

The historical schedules above acquire a records meaning. Marks placed at the cuts of Hui Shi's stick, however many, reach at most $\frac{6}{7}K_\tau/\kappa$, and the two quarter-cuts a day reach $\frac{27}{28}K_\tau/\kappa$: the stick that is "not exhausted in ten thousand generations" leaves one seventh of the discernibility of the comparison permanently out of reach, because the pieces it takes away are never marked again. If the zero-floor model supplies arbitrarily accurate balanced marks, three distinct times instead permit error tending to zero; a zero lower product bound alone does not guarantee that availability.

## Weyl lifting and polygon phases

Transverse coordinate $\hat{y}$, momentum $\hat{p}$, $[\hat{y}, \hat{p}] = i\hbar$, mass $m > 0$; the horizontal motion at speed $v$ is common to every history considered and cancels. A force history is an impulse density or measure $f$ on $[0, \tau]$ with Hamiltonian $H_f(t) = \hat{p}^2/2m - f(t)\hat{y}$. In the interaction picture of the free evolution $U_0$, write $W_f(t) = U_0(t)^\dagger U_f(t)$ and use the Weyl operators of the probabilistic note,

$$D(a,b) = \exp\left[\frac{i}{\hbar}(b\hat{y} - a\hat{p})\right], \qquad D(z_1)D(z_2) = e^{-i\omega(z_1,z_2)/2\hbar}D(z_1+z_2),$$

with $\omega(z_1,z_2)=a_1b_2-a_2b_1$. An impulse $J$ at time $t$ is $U_0(t)^\dagger e^{iJ\hat y/\hbar}U_0(t)=D(-Jt/m,J)$, since $U_0(t)^\dagger\hat y U_0(t)=\hat y+\hat p t/m$. The history's phase-plane path is

$$z_f(t)=\big(\alpha_f(t),\beta_f(t)\big)=\Big(-\frac{1}{m}\int_0^t s\,f(s)\,ds,\ \int_0^t f(s)\,ds\Big),$$

the transverse displacement relative to free motion referred back to $t=0$ by the free flow, and the accumulated impulse. For the constant force, $z_F(t)=(-Ft^2/2m,\ Ft)$, a parabola in the phase plane.

Lemma 1. $W_f(\tau)=\exp\big[\frac{i}{2\hbar}\int_0^\tau\omega(z_f,dz_f)\big]\,D\big(z_f(\tau)\big)$.

Proof. For finitely many impulses with increments $\delta z_1,\dots,\delta z_N$ in time order, induction on the Weyl relation gives $D(\delta z_N)\cdots D(\delta z_1)=\exp[\frac{i}{2\hbar}\sum_{i<j}\omega(\delta z_i,\delta z_j)]D(\sum_j\delta z_j)$, and $\sum_{i<j}\omega(\delta z_i,\delta z_j)=\sum_j\omega(z_{j-1},\delta z_j)$ with $z_{j-1}=\sum_{i<j}\delta z_i$. For a density $f$ the generator $\frac{i}{\hbar}f(t)(\hat y+\hat p t/m)$ has c-number commutators at different times, so the Magnus series stops at second order and the Riemann sums converge to $\int\omega(z_f,dz_f)$, as in N01 § 4. □

Lemma 2. If $z_f(\tau)=z_g(\tau)$, then $W_fW_g^\dagger=e^{i\mathcal{A}(f,g)/\hbar}\,1$, where $\mathcal{A}(f,g)=\frac12\oint\omega(z,dz)$ is the signed area enclosed by $z_f$ followed by $z_g$ reversed.

Proof. Lemma 1 for both, and Green's theorem for the closed loop. □

The specified linear-force histories with equal phase-plane increments induce the same endpoint canonical map on the body. Later observations of that body alone do not recover which force was used; an apparatus or memory can retain it. Lemma 2 says the quantum evolutions still differ, by the enclosed phase-plane area over $\hbar$, and that the difference is a multiple of the identity. This is the standard geometric phase of displaced oscillators, used in trapped-ion phase gates (Sørensen and Mølmer, PRA 62, 022311, 2000; Leibfried et al., Nature 422, 412, 2003, both metadata). The $t^3$ phase of a freely falling wave packet is the single-history form of it (Greenberger and Overhauser, RMP 51, 43, 1979, metadata).

## The inscribed polygon

Partition $0=t_0<\cdots<t_N=\tau$ with $\tau_j=t_j-t_{j-1}$ and midpoints $c_j=(t_{j-1}+t_j)/2$. Newton's inscribed polygon gives the impulse $F\tau_1/2$ at $t_0$, $F(\tau_j+\tau_{j+1})/2$ at each interior $t_j$, and $F\tau_N/2$ at $t_N$: each step's impulse $F\tau_j$ is split equally between its two ends.

Theorem P1. (i) On $(t_{j-1},t_j)$ the polygon moves uniformly with momentum $Fc_j$ along the chord of the parabola, and its phase-plane path passes through $z_F(t_j)$ after each completed cell. (ii) $W_{\text{poly}}(\tau)=e^{i\theta_N}W_F(\tau)$ with $\theta_N=F^2\sum_j\tau_j^3/(24m\hbar)$. (iii) With $S_j=vF\tau_j^3/(12m)$ the area between the $j$-th arc and its chord, $\theta_N=(F/2v\hbar)\sum_j S_j$.

Proof. (i) After the second half of the impulse at $t_{j-1}$ the momentum is $Ft_{j-1}+F\tau_j/2=Fc_j$, and in time $\tau_j$ it carries the body through $Fc_j\tau_j/m=F(t_j^2-t_{j-1}^2)/2m$, the parabola's transverse increment. In the phase plane the two half-impulses of step $j$ are $k_1=(-F\tau_jt_{j-1}/2m,\ F\tau_j/2)$ and $k_2=(-F\tau_jt_j/2m,\ F\tau_j/2)$, whose sum is the parabola's increment $\delta z_j=(-F\tau_jc_j/m,\ F\tau_j)$.

(ii) Both evolutions are products over steps, in the same order, of operators with the same displacements $\delta z_j$, so their ratio is the product of the per-step phase ratios. By Lemma 1 the polygon's step is $D(k_2)D(k_1)=e^{i\omega(k_1,k_2)/2\hbar}D(\delta z_j)$ with $\omega(k_1,k_2)=\frac{F\tau_j}{2}\cdot\frac{F\tau_j}{2m}(t_j-t_{j-1})=\frac{F^2\tau_j^3}{4m}$. The curve's step, with $s=t-t_{j-1}$ and path $(-F(s^2+2st_{j-1})/2m,\ Fs)$ from $z_F(t_{j-1})$, has $\omega(z,dz)=\frac{F^2s^2}{2m}ds$, hence phase $\frac{1}{2\hbar}\int_0^{\tau_j}\frac{F^2s^2}{2m}ds=\frac{F^2\tau_j^3}{12m\hbar}$. Neither depends on $t_{j-1}$. The ratio is $\exp[\frac{i}{\hbar}(\frac18-\frac{1}{12})\frac{F^2\tau_j^3}{m}]=\exp[\frac{iF^2\tau_j^3}{24m\hbar}]$.

(iii) Over the step, the chord exceeds the parabola by $\frac{F}{2m}(t-t_{j-1})(t_j-t)$, and integrating against $dx=v\,dt$ gives $S_j=vF\tau_j^3/12m$. Then $\frac{F}{2v}S_j=\frac{F^2\tau_j^3}{24m}$. □

Lemma XI's tangent area for the step is $A_j = vF\tau_j^3/6m = 2S_j$, so $\theta_N = (F/4v\hbar)\sum_j A_j$, and Proposition 1 of the paper is the statement that this sum vanishes with the mesh. In the phase plane the polygon's path is the circumscribed tangent polygon of the parabola $z_F$: each impulse at $t_j$ moves along the tangent direction $(-t_j/m, 1)$, and the two paths touch at every $z_F(t_j)$, which is why the area between them splits step by step. The phase is also the difference of the two histories' classical actions between the same endpoints, whatever the endpoints, which is why a comparison of quadratic Hamiltonians leaves only a global phase.

Corollary P2 (insertion). Replacing step $j$ by steps $\lambda\tau_j$ and $(1-\lambda)\tau_j$ lowers $\theta_N$ by

$$\frac{F^2\tau_j^3}{24m\hbar}\big[1-\lambda^3-(1-\lambda)^3\big] = \frac{F^2\tau_j^3\,\lambda(1-\lambda)}{8m\hbar} = \frac{F}{2v\hbar}\,T_j(\lambda),$$

with $T_j(\lambda) = S_j - S_j' - S_j''$ the triangle inscribed between the old chord and the new vertex. At $\lambda = \frac{1}{2}$, $T_j = \frac{3}{4}S_j$, which is Archimedes' Quadrature of the Parabola, Propositions 17 and 24 (the segment is four thirds of its inscribed triangle, by balancing and by exhaustion; companion). After $k$ rounds of halving every step, $\theta$ is $4^{-k}$ of its value. □

Each inserted point therefore changes the quantum evolution by a computable phase, fixed by the area the insertion removes, and the change is the same for every state of the body. A step's own phase reaches one radian at

$$\tau_\hbar = \left(\frac{24m\hbar}{F^2}\right)^{1/3},$$

the scale that the constant-force note found for the single chord lens. The insertion mesh of the mark theorem, $\tau_* = (48z_{1-\epsilon}^2 m\hbar/F^2)^{1/3}$, has the same form, obtained there from the cost of a mark and here from the phase of a segment with no apparatus at all.

## Archimedes' geometric series

The chord/arc area is $S_j = vF\tau_j^3/(12m)$; the tangent/arc area is $A_j = 2S_j$, and the tangent triangle is $3S_j$. Halving inserts the triangle $T_j = 3S_j/4$. Archimedes, Quadrature of the Parabola, Propositions 17, 23 and 24, proves the segment is four thirds of its inscribed triangle and sums the exhaustion series $T_j + T_j/4 + T_j/16 + \cdots = 4T_j/3 = S_j$ (Heath companion, passages read). Thus each halving removes one stage of the same energy series: $\theta_{\text{step}} = (F/2v\hbar)S_j$, and the total remainder falls by four each complete round. A one-radian comparison is a scale for this specified quantum phase experiment, not a prohibition on finer geometry.

The preface to The Method credits Democritus with asserting the cone's one-third volume and Eudoxus with its proof (source, passage). The Galileo area also has a one-third factor, $A_{\text{inertial,fall}} = vF\tau^3/(6m) = v\tau s/3$ for sagitta $s = F\tau^2/(2m)$. This modern geometric comparison establishes neither transmission nor a record-disturbance floor for a static solid.

## Ordering holonomy and the partition identity

Keep the curve's step operators $V_j = e^{i\varphi_j}D(\delta z_j)$ and compose them in another order. For a permutation $\sigma$, write $U_\sigma$ for the product with the steps applied in the order $\sigma$.

Theorem P3. $U_{\text{chron}}U_\sigma^\dagger = \exp[\frac{i}{\hbar}\sum\omega(\delta z_i, \delta z_j)]1$, the sum over the pairs $i<j$ that $\sigma$ inverts. Every term is positive,

$$\omega(\delta z_i, \delta z_j) = \frac{F^2}{m}\tau_i\tau_j(c_j - c_i) > 0 \qquad (i<j),$$

so the reversed order is extremal, with

$$\Phi_N = \frac{1}{\hbar}\sum_{i<j}\omega(\delta z_i,\delta z_j) = \frac{F^2}{6m\hbar}\Big(\tau^3 - \sum_j \tau_j^3\Big).$$

Proof. The per-step phases $\varphi_j$ are common to every ordering. By the Weyl relation the phase of a product is $\frac{1}{2\hbar}$ times the sum of $\omega(\delta z_a,\delta z_b)$ over pairs with $a$ applied before $b$; a pair kept in order cancels between the two products, and an inverted pair contributes $\omega(\delta z_i,\delta z_j)-\omega(\delta z_j,\delta z_i) = 2\omega(\delta z_i,\delta z_j)$. With $\delta z_i = (-F\tau_i c_i/m,\ F\tau_i)$, $\omega(\delta z_i,\delta z_j) = \frac{F^2}{m}\tau_i\tau_j(c_j-c_i)$. For the sum, $c_j - c_i = \frac{\tau_i}{2} + \sum_{i<k<j}\tau_k + \frac{\tau_j}{2}$, so $\sum_{i<j}\tau_i\tau_j(c_j-c_i) = \frac{1}{2}\sum_{i\neq j}\tau_i^2\tau_j + \sum_{i<k<j}\tau_i\tau_k\tau_j$, and $(\sum_j\tau_j)^3 = \sum_j\tau_j^3 + 3\sum_{i\neq j}\tau_i^2\tau_j + 6\sum_{i<k<j}\tau_i\tau_k\tau_j$ gives $\frac{1}{6}(\tau^3 - \sum_j\tau_j^3)$. □

Corollary P4 (a partition identity). $\Phi_N + 4\theta_N = F^2\tau^3/(6m\hbar)$ for every partition, and $F^2\tau^3/6m = \tau\Delta E/3 = (F/v)A_{\text{inertial,fall}}$ with $A_{\text{inertial,fall}} = vF\tau^3/6m$ the paper's area (1). □

In areas, $\hbar\Phi_N = (F/v)(A - \sum_j A_j)$ and $4\hbar\theta_N = (F/v)\sum_j A_j$: the ordering holonomy carries the part of the Galileo area that refinement keeps, and the polygon defect carries the part that Proposition 1 sends to zero. For the uniform partition, $\Phi_N = \frac{\tau\Delta E}{3\hbar}(1-N^{-2})$.

The chronological and reversed products realize the same displacement, so the holonomy is a property of the composition law. No force history acting on a body realizes a reordering other than the chronological one: an impulse applied at the wrong time must be translated back, so a reversed order needs displacements as well as impulses, which trapped-ion practice supplies. Theorem P3 is therefore a statement about the composition law, and Theorem P1 is the one that needs only two force histories acting on a real body.

### Controlled readout and finite resources

Put the two histories of Theorem P1 on the arms of a control qubit, as in N01 § 4. The body ends in the same state on both arms, so the qubit acquires the relative phase $\theta_N$ and nothing else, and the optimal single-shot error in deciding polygon against curve is $\epsilon = \frac{1}{2}(1 - |\sin(\theta_N/2)|)$, with N01's formula for $n$ copies. In one controlled pass, a single step is resolved from its chord at error $\epsilon$ only if

$$\frac{F^2\tau_j^3}{24m\hbar} \ \geq\ 2\arcsin(1-2\epsilon), \qquad \tau_j \ \geq\ \left(\frac{48\,m\hbar\ \arcsin(1-2\epsilon)}{F^2}\right)^{1/3},$$

for every state of the body, pure or mixed, squeezed or not. The bound concerns this readout and its resource, the number of controlled passes: $k$ passes accumulate $k\theta_N$ and lower the threshold on $\tau_j$ by $k^{1/3}$, while independent copies obey the separate Helstrom formula in the insertion note. The body's own displacement is a separate record, priced by the mark and aperture theorems, and squeezing acts on that record. What the phase supplies is the part of the signal that no preparation of the body can move.

## The fits: a conditional historical action count

The Opticks Book II Part III Proposition XVII makes the fit interval in two media proportional to the sine-of-incidence/refraction ratio; Proposition X posits the inverse ratio for corpuscle speeds in Newton's emission theory. At fixed corpuscle mass this gives a per-colour $\Lambda v$, hence $\Lambda p$, invariant under that comparison. The held passages are source evidence for that historical reading. Its wavelength analogy was already

discussed by Sakkopoulos 1988 (metadata), Whittaker and de Broglie's 1929 Nobel lecture. No novelty is claimed here.

This interval is a longitudinal periodic disposition. It is not a transverse position uncertainty or a conjugate momentum spread. Extending its rule to a continuously varying speed is an additional hypothesis, stated in the next proposition, rather than a theorem Newton printed.

Proposition F5. Suppose, extending Props. X and XVII from a step in the medium to a continuously varying speed, that a corpuscle's fits advance by one interval $\Lambda = \Lambda_0 p_0/|p|$ per length of path. Then the number of fits along a path is its Maupertuis action $W = \int p \cdot dq$ in units of $\Lambda_0 p_0$. For the inscribed polygon and the curve of Theorem P1 the counts differ by

$$\frac{\Delta W}{\Lambda_0 p_0} = \frac{F^2}{12 m \Lambda_0 p_0} \sum_j \tau_j^3 = \frac{1}{2\Lambda_0 p_0} \frac{F}{v} \sum_j A_j$$

fits, with $A_j$ Lemma XI's tangent areas.

Proof. Along the actual motion $p \parallel dq$, so $|p|\, ds = p \cdot dq$ and the count is $\int p \cdot dq/(\Lambda_0 p_0)$. The horizontal parts are common. On step $j$ the curve has $\int p_y\, dy = \frac{F^2}{m} \int_{t_{j-1}}^{t_j} t^2\, dt$ and the chord has $\frac{F^2}{m} c_j^2 \tau_j$; their difference is $\frac{F^2}{m} \tau_j \big[\frac{1}{3}(t_{j-1}^2 + t_{j-1} t_j + t_j^2) - \frac{1}{4}(t_{j-1} + t_j)^2\big] = \frac{F^2 \tau_j^3}{12m}$, and $F^2 \tau_j^3/6m = (F/v) A_j$. □

The count is a classical internal-phase variable attached to a trajectory. Proposition XII's determinate reflection/transmission disposition and Query 29's overtaking vibrations do not supply Hilbert states, Born probabilities, Weyl relations or coherent addition of amplitudes. Proposition XV's oblique-emergence rule also need not be a path-length count. Even where the conditional action count is useful, it does not establish a quantum central extension.

If one additionally sets $\Lambda = \lambda/2$ and $\lambda p = 2\pi\hbar$, counting $2\pi$ per fit gives $4\theta_N$ for this pair of histories: one factor two is the half-wavelength interval, the other is $\Delta W = 2\Delta S$. The wave phase advancing $\pi$ per interval gives $2\theta_N$. These are conditional comparisons using quantum calibration. A kick $J$ before a displacement $d$ changes $\int p\, dq$ by $Jd$ relative to reversing their order; a classical phase count can record this area without becoming a quantum composition law. Universality across colours and systems, a positive physical constant, and the quantum state/instrument rules remain separate physical assumptions.

## General force histories: one Dirichlet functional

Let $f$ be any integrable force history on an interval of duration $\tau$, and compare the motion with its chord, the uniform motion between the same two positions. The deviation $\delta y = y_{\text{chord}} - y_f$ obeys $m\, \delta y'' = -f$ with $\delta y = 0$ at both ends, so $\delta y = G_\tau f/m$ with the Dirichlet Green's function $G_\tau(s,u) = \min(s,u)\,(\tau - \max(s,u))/\tau$ in time measured from the start of the interval. Define the kinetic action of the deviation

$$\mathcal{K}_\tau[f] = \int_0^\tau \tfrac{1}{2} m\, \delta\dot{y}^{\,2}\, dt = \frac{1}{2m} \int_0^\tau \int_0^\tau G_\tau(s,u) f(s) f(u)\, ds\, du \;\geq 0,$$

zero only for $f = 0$. For the constant force it is $F^2\tau^3/24m = \tau \Delta E/12 = (F/2v)S$, with $S$ the segment of the chord.

Theorem G7. (a) On each step of a partition, the two impulses $J_j^- = \int (1 - s/\tau_j) f$ and $J_j^+ = \int (s/\tau_j) f$ at its ends reproduce the step's impulse and displacement, so Newton's polygon built from them agrees with the motion at every vertex, and

$$W_{\text{poly}} = \exp\Big[\frac{i}{\hbar}\sum_j \mathcal{K}_{\tau_j}[f]\Big] W_f.$$

Inserting vertices never raises the phase: $\sum_j \mathcal{K}_{\tau_j}[f] \le \mathcal{K}_\tau[f]$. (b) For every protocol of Gaussian marks with uncorrelated error and recoil and $\delta_j\Delta_j \ge \kappa$, with the initial position and velocity unknown, the deflection between the motion under $f$ and free motion obeys

$$d^2 \;\le\; \frac{\mathcal{K}_\tau[f]}{\kappa},$$

with equality approached by dense protocols when $f$ has one sign. (c) Hence at $\kappa = \hbar/2$ the best deflection any such protocol can reach is twice the phase between the motion and its chord when $f$ has one sign, $d^2_{\max} = 2\,\mathcal{K}_\tau/\hbar = 2\theta_1$.

Proof. (a) Measure time from the start of the step. The impulses at 0 and $\tau_j$ are $k^- = (0, J^-)$ and $k^+ = (-\tau_j J^+/m,\ J^+)$ in the phase plane, and $J^- + J^+ = \int f$, $\tau_j J^+ = \int sf$ match the step's increment. Lemma 1 gives the polygon's step phase $\frac{1}{2\hbar}\omega(k^-, k^+) = \tau_j J^+ J^-/2m\hbar$ and the motion's step phase $\frac{1}{2\hbar}\int \omega(z, dz) = \frac{1}{2m\hbar}\iint_{u<s}(s-u)f(s)f(u)$, because along the path $\omega(z, dz) = f(s)\,\frac{1}{m}\int_0^s (s-u)f(u)\,du\,ds$. Now $\tau_j J^+ J^- = \iint s(\tau_j - u)f(s)f(u)/\tau_j$; symmetrizing and subtracting $\frac{1}{2}\iint |s-u| ff$ leaves the kernel $\frac{1}{2\tau_j}[s(\tau_j - u) + u(\tau_j - s) - \tau_j|s-u|] = G_{\tau_j}(s,u)$. Shifting the time origin is the shear $(a,b) \mapsto (a - t_{j-1}b/m, b)$, which preserves $\omega$, so the per-step phases are $\mathcal{K}_{\tau_j}/\hbar$, and they multiply as in Theorem P1. The Dirichlet principle $2m\mathcal{K}_\tau = \sup_T(2\int fT - \int T'^2)$ over $T$ vanishing at the ends, restricted to $T$ vanishing also at the interior vertices, gives the monotonicity.

(b) As in Theorem 2 of the paper, $u^{\mathsf{T}}\Sigma u \ge 2\kappa E/m$ and $u^{\mathsf{T}}P = \frac{1}{m}\int_0^\tau fT$ with $T$ vanishing at both ends. Since $-(G_\tau f)'' = f$, $\int fT = \int (G_\tau f)'T' \le \sqrt{\langle f, G_\tau f\rangle}\sqrt{E}$, so $d^2 \le \langle f, G_\tau f\rangle/(2m\kappa) = \mathcal{K}_\tau/\kappa$. For $f$ of one sign take $T \propto G_\tau f$, which has one sign while $T''$ has the other, so the interior weights and the $S_j$ have opposite signs throughout and the dense balanced construction of Theorem 2 applies.

(c) Theorem P1 with a single step is the chord, whose phase is $\theta_1 = \mathcal{K}_\tau/\hbar$. Equality of the optimized mark information uses the one-sign attainability condition in (b); for sign-changing $f$ only the upper bound is asserted. □

The two legs of the programme are governed by one functional of the force history. $\mathcal{K}_\tau$ is what Newton's refinement removes: it is the kinetic action of the motion relative to its chord, and for a constant force it is the chord's segment times $F/2v$. Planck's constant converts it into the phase by which the motion differs from its inscribed polygon, a c-number that no preparation of the body moves; the mark cost converts it into the largest statistical distance any record of the motion can reach within the Gaussian model of marks. At $\kappa = \hbar/2$ the optimized second quantity is exactly twice the first for the one-sign force family; the general statement is an upper bound. The two legs stand on different footing: the phase is exact and uses the Weyl relations alone, while the mark bound holds inside a linear Gaussian model with the trade-off $\delta\Delta \ge \kappa$, a model that squeezing and grid probes leave, as the recoil note records. The 24 in Theorem P1's $F^2\tau^3/24m$ and in the sharp bound $d^2 \le F^2\tau^3/24m\kappa$ of the paper's Theorem 2 is one integral, $\iint G_\tau = \tau^3/12$, multiplied by $F^2/2m$.

## Consequence for STATE

Cut energy, Weyl phase and Gaussian information share an exact functional, while their physical premises remain distinct. A cut-share floor differs from a cell-action

floor by $3s(1-s)$; controlled-pass resources and Gaussian-mark availability must accompany any resolution claim. Newton's fits supply source evidence for a conditional classical phase count, not a derivation of the Weyl relations or physical $\hbar$. The independent positive-action question remains open. Historical cut schedules retain their own scripts and edition links; their convergence is not evidence of transmission or of least magnitudes.